Problem 1

$$f(x,y) = x - y, \qquad R = [0,2] \times [0,1].$$

Let V be the volume between $z = f(x,y)$ and $z = 0$ over R.

1. Evaluate $J = \iint_R f(x,y)\,dA$.
2. Sketch the parts R_+ and R_- of R where $f \geq 0$ and $f \leq 0$, respectively. Find both areas using double integrals.
3. Find the volumes V_+ and V_- between the two planes over R_+ and R_-, respectively.
4. Use part (3) to find V and the average value of $|f|$ on R.

Problem 2

For (1)–(2): $$I = \int_0^1 \int_{\sqrt{x}}^1 e^{y^3}\, dy\, dx.$$

1. Sketch the region of integration for I, and change the order of integration.
2. Evaluate I using the reversed order.

For (3)–(4): $$E = \{(x, y) : y \geq 0,\ x^2 + y^2 \leq 2x\}.$$

3. Sketch E, give its polar-coordinate bounds, and find its area using a double integral.
4. Use polar coordinates to find the volume under $z = y$ and above E.